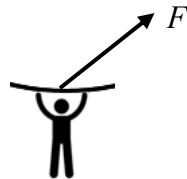


Q2(a)



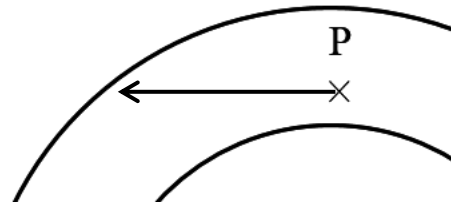
Q2(b)

The ceiling exerts a tangential force on the astronaut to increase the magnitude of his tangential velocity (tangential acceleration).

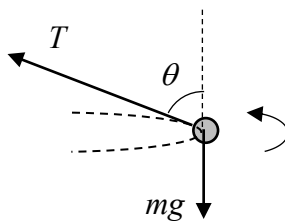
The ceiling also exerts a radial force on him to change the direction of his velocity (centripetal acceleration).

The resultant of the tangential and radial forces is a force that is inclined with respect to the radius.

Q2(c)



Q2(d)



Let g = apparent acceleration (due to circular motion).

Net vertical force = 0

$$T \cos \theta = mg \quad \dots \dots (1)$$

Centripetal force = horizontal component of tension

$$mr\omega^2 = T \sin \theta$$

$$m(L \sin \theta)\omega^2 = T \sin \theta$$

$$\therefore mL\omega^2 = T$$

$$mL\omega^2 = \frac{mg}{\cos \theta}, \text{ using (1).}$$

$$g = L\omega^2 \cos \theta$$

$$g = 0.958 \times \left(\frac{2\pi}{1.185} \right)^2 \times \cos 65.2^\circ$$

$$\therefore g = 11.297$$

$$\therefore g = \mathbf{11.3 \text{ m s}^{-2}}.$$

Q2(e)

The acceleration found in (d) is the centripetal acceleration.

$$\text{Centripetal force} = ma_c = mr\omega^2$$

$$mg = mr\omega^2 \text{ or } g = r\omega^2$$

$$\therefore 11.3 = r \times \left(\frac{2\pi}{14.2} \right)^2$$

$$\therefore \text{radius of circular motion of astronaut} = r = \mathbf{57.7 \text{ m}}$$

$$\Omega^2(r)$$